# Practice workbook · Drawing the lines for an inequality graph

Drawing the straight boundary lines of a set of linear inequalities accurately on a grid or axes.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-05

## 1. Question 5 · part a · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part a only.

**5** (a) On the grid opposite draw the line with equation

(i) $y = 2x + 5$ (ii) $4y = x - 8$ (iii) $5y + 3x = 30$

**(3)**

(b) Show, by shading, the region $R$ defined by the inequalities

$y \leqslant 2x + 5$ $4y \geqslant x - 8$ $5y + 3x \leqslant 30$

**(1)**

For all points in $R$ with coordinates $(x, y)$

$P = 2x - 5y$

(c) Using your graph, find the least value of $P$

**(3)**

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## 1. Question 5 · part a · 3 marks · Working and answer

Working and answer

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## 2. Question 2 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

**2** (a) Using the axes below sketch the line with equation

(i) $y = 6$ (ii) $y + x = 10$ (iii) $y = 2x - 5$

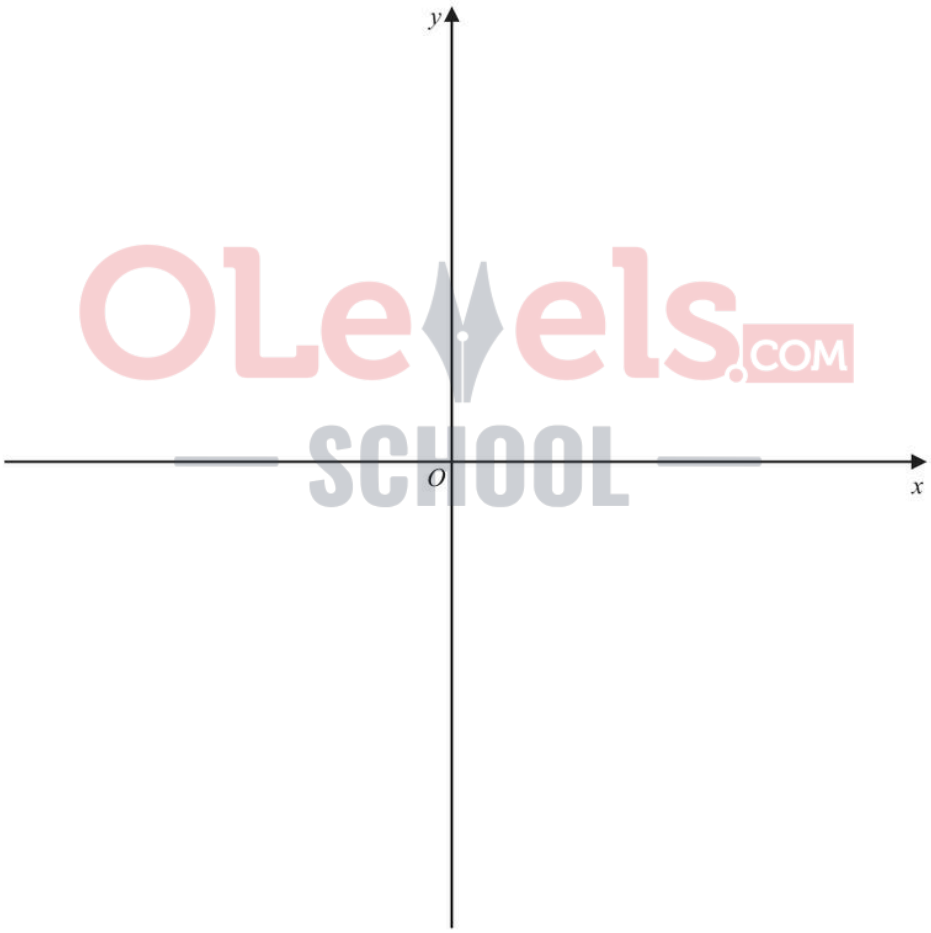
Show the coordinates of any point where each line crosses the coordinate axes.

**(3)**

(b) Show, by shading on your sketch, the region $R$ defined by the inequalities

$y \leqslant 6$ $y + x \leqslant 10$ $y \geqslant 2x - 5$ $x \geqslant 0$

**(1)**




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## 2. Question 2 · part a · 3 marks · Working and answer

Working and answer

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## 3. Question 4 · part a.i · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part a.i only.

**4** (a) On the grid opposite, draw the line with equation

(i) $y = 2x - 4$ (ii) $2x + 3y = 12$ (iii) $y + 2x + 2 = 0$

**(3)**

(b) Show, by shading on the grid, the region $R$ defined by the inequalities

$y \geqslant 2x - 4$ $2x + 3y \leqslant 12$ $y + 2x + 2 \geqslant 0$

**(1)**

For all points in $R$, with coordinates $(x, y)$

$$P = x - 2y$$

(c) find the least value of $P$

**(4)**

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## 3. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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# 4. Question 4 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

**4** (a) On the axes opposite, draw the line with equation

(i) $y = -x - 1$ (ii) $y - 3x + 8 = 0$ (iii) $2y = x + 8$

**(3)**

(b) Show, by shading on your graph, the region $R$ defined by the inequalities

$y \geqslant -x - 1$ and $y \geqslant 3x - 8$ and $2y \leqslant x + 8$

**(1)**

For all points in $R$, with coordinates $(x, y)$

$$P = 2y - 3x$$

(c) Find

(i) the greatest value of $P$

(ii) the least value of $P$

**(4)**

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P 7 3 5 8 6 A 0 8 3 2

## 4. Question 4 · part a · 3 marks · Working and answer

Working and answer

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## 5. Question 4 · part a · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

**4** (a) On the grid opposite, draw the graph of the line with equation

(i) $3x+2y=18$ (ii) $x+3y-6=0$ (iii) $y=3x$

**(3)**

(b) Show, by shading on the grid, the region $R$ defined by the inequalities

$3x+2y \leqslant 18$ $x+3y-6 \geqslant 0$ $y \leqslant 3x$

**(1)**

For all points in $R$, with coordinates $(x, y)$

$$P=2x-y$$

(c) find the least value of $P$

**(4)**

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## 5. Question 4 · part a · 3 marks · Working and answer

Working and answer

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## 6. Question 5 · part a · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

**5** (a) On the grid opposite, draw the graphs of the lines with equations

$$2x + 3y = 24 \qquad y = 2x \qquad 3y = 2x - 12$$

**(3)**

(b) Show, by shading on the grid, the region $R$ defined by the inequalities

$$2x + 3y \leqslant 24 \qquad y \leqslant 2x \qquad 3y \geqslant 2x - 12 \qquad y \geqslant 0$$

**(1)**

For all points in $R$, with coordinates $(x, y)$

$$F = 2x + 5y$$

(c) Find the greatest value of $F$.

**(3)**

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## 6. Question 5 · part a · 3 marks · Working and answer

Working and answer

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## 7. Question 2 · part a · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part a only.

**2** (a) On the grid below, draw the line with equation

(i) $5x + 2y = 10$ (ii) $y = x$

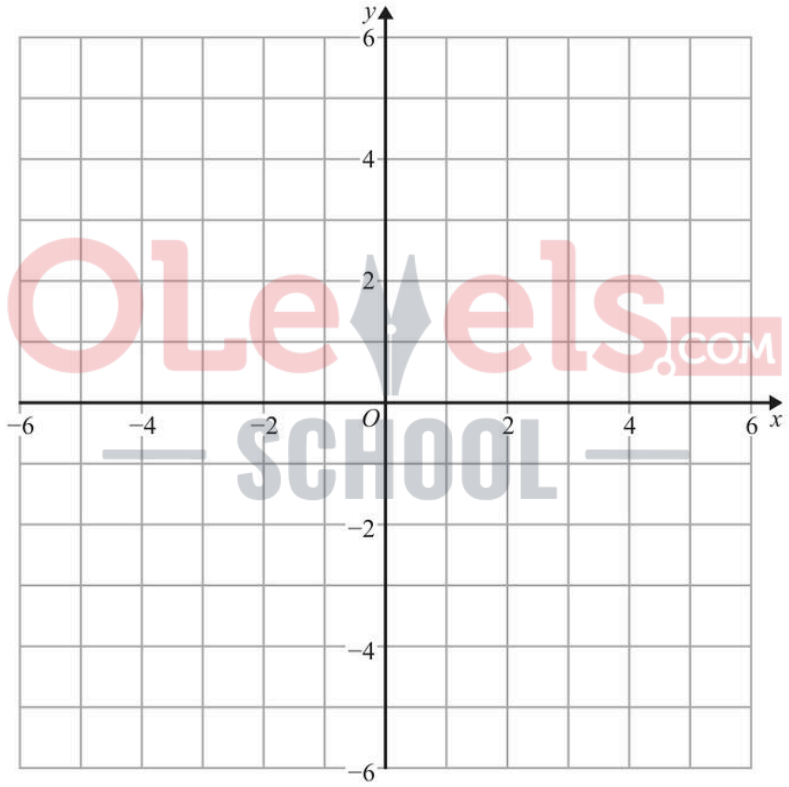
**(2)**

(b) Show, by shading on the grid, the region $R$ defined by the inequalities

$y \leqslant x$ $5x + 2y \leqslant 10$ $y \geqslant -2$ $x \geqslant 1$

Label the region $R$.

**(2)**




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P 5 9 9 3 8 A 0 6 3 6

## 7. Question 2 · part a · 2 marks · Working and answer

Working and answer

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## 8. Question 1 · part a.i · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part a.i only.

**1** (a) On the grid below, draw the line with equation

(i) $3x+4y=24$ (ii) $2x-5y+10=0$

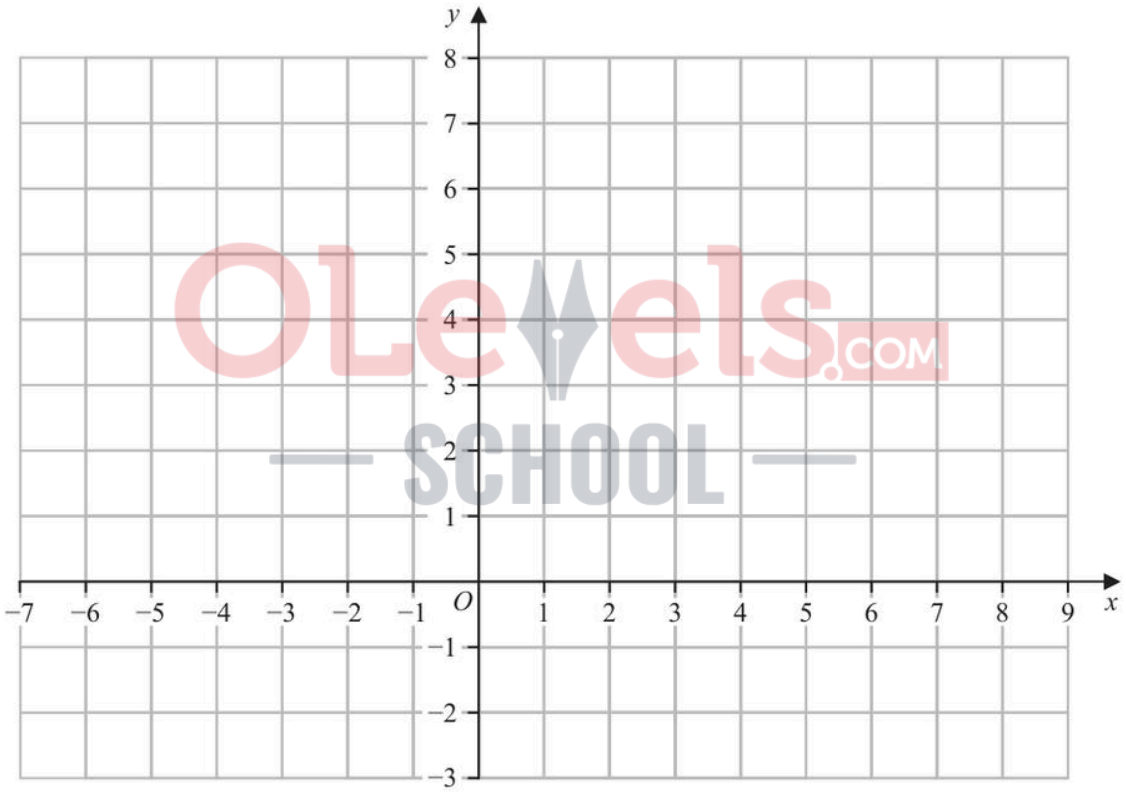
**(2)**

(b) Show, by shading on the grid, the region $R$ defined by the inequalities

$3x+4y \leqslant 24$ $2x-5y+10 \geqslant 0$ $y \leqslant 5$ $x \geqslant -1$

Label the region $R$

**(2)**




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## 8. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

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## 9. Question 1 · part a · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

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**1** (a) On the grid below, draw the graph of the line with equation

(i) $4x + 5y = 20$ (ii) $3y - 4x = -12$

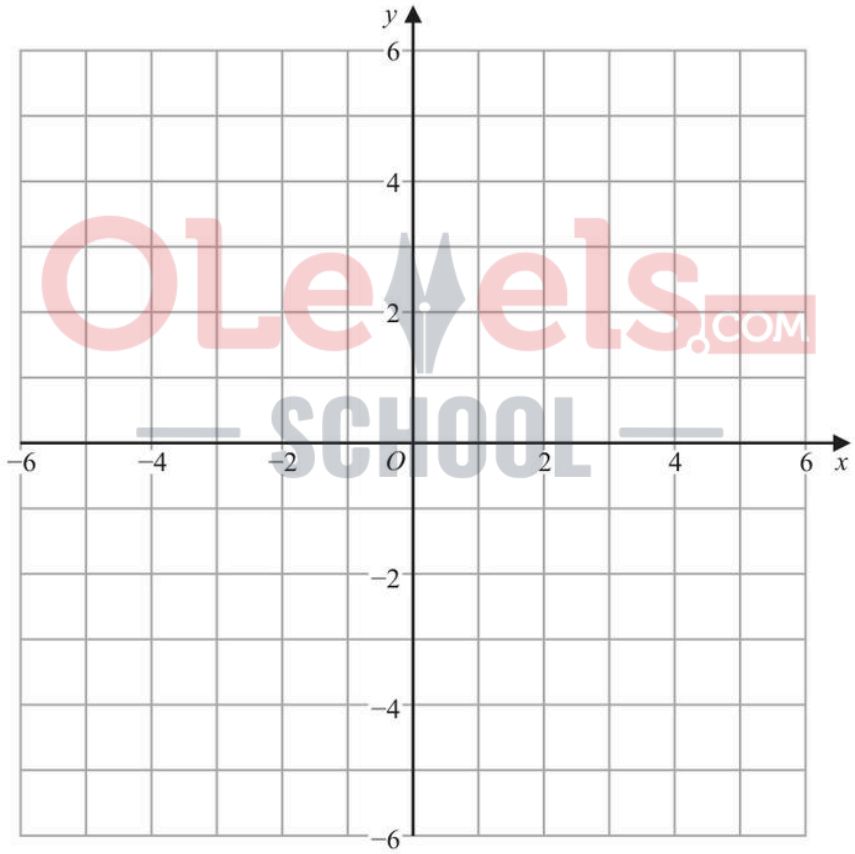
**(2)**

(b) Show, by shading on the grid, the region $R$ defined by the inequalities

$4x + 5y \leqslant 20$ $3y - 4x \geqslant -12$ $y \leqslant 3$ $x \geqslant 1$

Label the region $R$.

**(2)**




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**Turn over for a spare grid if you need to redraw your graph.**

## 9. Question 1 · part a · 2 marks · Working and answer

Working and answer

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## 10. Question 5 · part a.i · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a.i only.

**5** (a) On the grid opposite, draw the line with equation

(i) $3x-4y=12$ (ii) $y+6+3x=0$ (iii) $3y=18-x$

**(3)**

(b) Show, by shading on the grid, the region $R$ defined by the inequalities

$3x-4y \leqslant 12$ $y+6+3x \geqslant 0$ $3y \leqslant 18-x$

**(1)**

For all points in $R$, with coordinates $(x, y)$

$$P=3x-2y$$

Using values from your graph,

(c) find the least value of $P$ and the greatest value of $P$

**(4)**

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## 10. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

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